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[Paper 7] Paper "Metallicity" - 1e12q Hydro Run to Study Metals

This page presents the outline, ongoing efforts, results, and future plans for the AGORA Paper "Metallicity".

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Recent changes that require the most attention are in red color.

(last revision: 08/15/2018)

[1. Project Outline](#)

	Paper "Metallicity"										
GOALS	Connect galaxy evolution and LIGO observations.										
LEADERS	(• Sukanya Chakrabarti (Rochester IT) – proposal submitter)										
PARTICIPANTS	- Interested code representatives: <table border="1"> <thead> <tr> <th>Code</th><th>Contacts</th></tr> </thead> <tbody> <tr> <td>GADGET/GIZMO</td><td>Chakrabarti, Lewis</td></tr> <tr> <td>GEAR</td><td>(Revaz)</td></tr> <tr> <td>RAMSES</td><td>(Pallottini, Teyssier)</td></tr> <tr> <td>[to be added]</td><td></td></tr> </tbody> </table> (Butler may be interested in running tests using an isolated disk IC with ENZO, in connection with Paper "Disk-II"; see Section 3 below / Analysis help: Kim) - Working Groups engaged: - IX (Characteristics of Cosmological Galaxies), - XII (Interstellar Medium)	Code	Contacts	GADGET/GIZMO	Chakrabarti, Lewis	GEAR	(Revaz)	RAMSES	(Pallottini, Teyssier)	[to be added]	
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SCIENCE CASE	Belczynski et al. 2016 recently showed that the LIGO detection of GW150914 could be explained by the merger of massive, low-metallicity binary stars. A question that remains is what kind of galaxies would these progenitors be typically found in? Low-metallicity dwarf galaxies? - Initial work on understanding the host galaxies of LIGO events was done by O'Shaughnessy et al. 2016, showing that dwarf galaxies may contribute significantly to the LIGO signal, due to their low metallicity. Sufficiently low metallicity environments may also be found in the outer disks of spiral galaxies, which have yet to be studied. - The AGORA infrastructure is ideal for testing the effects of feedback on metallicity. It is already clear from Kim et al. 2016 that various solvers have some effect on the metallicity. Different kinds of feedback models are also likely to affect the metallicity in galaxy disks, and given the strong dependence on metallicity on LIGO detections, this in turn is likely to determine the most probable host galaxies for BH-BH mergers. - Regarding connecting LIGO to galaxy evolution, there's a possible connection using chemical evolution within galaxies (e.g., Cote et al. 2017). Interesting opportunities may exist to connect this to AGORA, given that some of the most interesting questions about NS-NS mergers come from the behaviors of ultra-faint dwarf galaxies (e.g., the recent Reticulum II work by Ji et al. 2016 and others). <i>[thoughts by O'Shea]</i> - Metal diffusion : comparison between different codes/methods <i>[thoughts by Revaz]</i> - Connecting galactic metal distributions and LIGO detection rates may contain lots										

	of uncertainties – such as star formation and metal production model, metallicity-dependent IMF, low metallicity cutoff, binary fraction, merging timescale, just to name a few. Therefore, let us not make our Paper "Metallicity" directly associated with LIGO signals upfront. • Comparing metal distribution in different simulations in an isolated setup could be extremely useful to compare metal diffusion in AMR vs. SPH vs. SPH+turbulent metal diffusion schemes (e.g., Shen et al.), metal transport, galactic metallicity gradient, star formation sites, bimodal metal content in globular clusters, etc. [thoughts by Abel/Kim] --> Kim and others are taking on this small project as of 08/2018.														
PRE- PRODUCTION DISCUSSION 1ST MEETING (04/24/2017) 2ND MEETING (08/12/2017) 3RD MEETING (08/11/2018)	• Pre-production discussion/to-do's: <table> <tr> <th>Item</th><th>To-do</th></tr> <tr> <td>Initial condition</td><td> • 10^{12} Ms halo at $z=0$ • We will piggyback on the planned runs for Paper "CGM" (CGM Workspace) and evolve them further down to $z=0$. In other word, this paper will be one of the causes to push the CGM run to lower redshift. </td></tr> <tr> <td>Common physics</td><td> • Similar to the CGM run (CGM Workspace). Please see their Workspace to learn about their planned calibration strategies for stellar feedback using an isolated disk run. • In addition to the calibration methods proposed for the Paper "CGM" we advocate to calibrate also on velocity dispersion as well as star formation rate. We also suggest to use an extended HI disk in the isolated run (to be decided). </td></tr> <tr> <td>Target redshift</td><td>• $z=0$</td></tr> <tr> <td>Output strategy</td><td>• See this link.</td></tr> <tr> <td>Common analysis</td><td> • We may begin from metallicity analysis routines in agora-analysis-script used for our disk comparison paper (formerly Paper "4") and build upon it. </td></tr> <tr> <td>Obs. counterpart</td><td> • e.g., Advanced LIGO constraints/predictions in the local Universe (to be determined) • Chakrabarti will provide more information on available observations related to metal distribution. </td></tr> </table> • Chakrabarti et al. have started a discussion, and to plan test problems using isolated ICs.	Item	To-do	Initial condition	• 10^{12} Ms halo at $z=0$ • We will piggyback on the planned runs for Paper "CGM" (CGM Workspace) and evolve them further down to $z=0$. In other word, this paper will be one of the causes to push the CGM run to lower redshift.	Common physics	• Similar to the CGM run (CGM Workspace). Please see their Workspace to learn about their planned calibration strategies for stellar feedback using an isolated disk run. • In addition to the calibration methods proposed for the Paper "CGM" we advocate to calibrate also on velocity dispersion as well as star formation rate. We also suggest to use an extended HI disk in the isolated run (to be decided).	Target redshift	• $z=0$	Output strategy	• See this link .	Common analysis	• We may begin from metallicity analysis routines in agora-analysis-script used for our disk comparison paper (formerly Paper "4") and build upon it.	Obs. counterpart	• e.g., Advanced LIGO constraints/predictions in the local Universe (to be determined) • Chakrabarti will provide more information on available observations related to metal distribution.
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OTHER LINKS	• WG IX Workspace (Characteristics of Cosmological Galaxies) • WG XII Workspace (Interstellar Medium)														

2. Results

(1) Calibration of Feedback Prescriptions

Code	Data Locations, Visualizations, Metrics to Compare

